



The implementation of green supply chain management (GSCM) and environmental management system (EMS) practices and its impact on market competitiveness during COVID-19

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Abstract

Despite great academic interest in global green supply chain management (GSCM) practices, its effectiveness for environmental management systems (EMS) and market competitiveness during COVID-19 remains untapped. Existing literature suggests that a fundamental link between GSCM, EMS, and market competitiveness is missing, as supply management is critical to maintain market competitiveness. To fill this gap in the literature, this study examines whether environmental management systems influence the link between GSCM practice and market competitiveness in China. We also propose the articulating role of big data analytics and artificial intelligence (BDA-AI) and environmental visibility toward these associations in the context of the COVID-19 pandemic. We evaluated the proposed model using regression-based structural equation modeling (SEM) with primary data ($n = 330$). This result provides empirical evidence of the impact of GSCM on EMS and market competitiveness. Moreover, the results show that the BDA-AI and the environmental visibility enhanced the positive relationship between GSCM-EMS and EMS and market competitiveness in China. Recent research shows that supply chain professionals, policymakers, managers, and researchers are turning to formal EMS, BDA-AI, and environmental visibility to help their organizations achieve the competitiveness that the market indicates they need.

Keywords Green supply chain management (GSCM) · Environmental management system (EMS) · Environmental visibility (EV) · Big data analytics and artificial intelligence (BDA-AI) · Market competitiveness (MC) · COVID-19

Introduction

International organizations are dedicated to promoting sustainable development through actions like stepping up green marketing, implementing agile systems, and

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creating a greener environment in order to comply with international requirements, as well as through introducing industry innovation and increasing environmental protection practices (like ISO-14001 certification in 2020). We might try to use potential developments to obtain a competitive advantage. A policy for addressing customer concerns about environmental issues reduces the negative effects of service and production doings on the environment (Cankurtaran and Beverland 2020). Organizations are being pushed into an era of uncertainty rather than competitiveness by a surge of megatrends like globalization, new technology, environmental sustainability challenges, and stakeholder relationships (Ahmed et al. 2021). The organizational climate, however, might be unstable, particularly during unforeseen events like the COVID-19 epidemic, which was reported in March 2020 and spread confusion and terror. During the terrible COVID-19 incident, small and medium-sized businesses (SMEs) struggle to develop survival plans, which include careful integration and consolidation to manage customer and supplier expectations and coordinate supply chain management (SCM) (Cankurtaran and Beverland 2020; Baierle et al. 2022). The COVID-19 breakout, which was swift and unexpected, severely interrupted the organization's SCM as a result, making the company highly unstable and suffer enormous losses quickly (Canhoto and Wei 2021). Additionally, social media and information and communication technology (ICT) growth have caused a rapid rise in environmental consciousness; this is another challenge businesses must overcome in a competitive market (Luqman et al. 2021).

Prior to estimating the COVID-19 pandemic's significant effects on businesses, it is important to consider how well the severe acute respiratory syndrome (SARS) outbreak of 2003 was managed. One nation, namely, China, is responsible for a \$3 billion to \$100 billion loss to the world economy (Wilder-Smith 2006; Luqman, Talwar et al. 2021). In contrast, the COVID-19, also referred to as "one-in-a-hundred-years pathogen" has shocked the world and is predicted to have a more significant and long-lasting effect on the global economy (Gates 2020). As a point of attention during this time, developing markets began to emerge, which reduced market returns and completely disrupted the world's financial markets (Cankurtaran and Beverland 2020). The quick dissemination of knowledge about the pandemic and the economic unpredictability in a connected world are what are driving the increased number of COVID-19-positive cases (Al-Awadhi et al. 2020). As a result, the interaction of these variables led to unprecedented outcomes, raising concerns among the international community about the organization's SCM during the pandemic (Baker et al. 2020). Businesses might not be able to exit the market despite this volatility since they require a diversity of supply chain and services to utilize their investable surpluses effectively (Talwar et al.

2021). In addition, if they are forced to stay at home during COVID-19 lockdowns and have a lot of recreational time, making the simple change to a greener and more maintainable environment will draw in more customers and investors to their business. We think it might be able to keep and draw investment (Bhatia and Gangwani 2021). An online interface was organized by big data and information communication technology to support SCM to provide sustenance toward the organizations (Maheshwari et al. 2021) and supports companies through online interfaces. However, as we observed during the COVID-19 pandemic, an organization's SCM is riskier in such a turbulent market and stakeholders may sustain significant losses. Therefore, it seems safer and more appealing to choose not to give goods or services rather than transporting goods to a quarantined area (Barrafreem et al. 2020). Based on the conversation, we think that when confronted with extraordinary occurrences that even the most knowledgeable corporation cannot understand, organizations may need to be shielded from their own bad judgment. In our perspective, it is essential to comprehend the SCM factors that influence an organization's competitiveness throughout the COVID-19 epidemic in China.

We believe that despite external constraints, the outbreak presents a rare chance for us to learn about green supply chain management (GSCM). These perceptions can be used to create regulations that secure organizations and better inform organizations about the dangers associated with choosing environmental actions in the face of unforeseeable situations like COVID-19 in China. To our information, the impact and repercussions of GSCM in the COVID-19 issue have not received much attention in prior studies. It is essential to research the GSCM outcomes as a result. With regard to tackling environmental challenges and enhancing organizational sustainability to achieve a competitive advantage, GSCM has recently attracted significant scholarly interest (Qader et al. 2022). The more sustainable supply process known as GSCM increases customer satisfaction, integrates and coordinates company activities, decreases supply chain waste, and lowers overall total cost of operations (Bhatia and Gangwani 2021) (Junaid et al. 2022). The ability of a company to locate, utilize, and incorporate both internal and external resources and information in order to facilitate all of its supply chain activities is known as supply chain capacity (Jiang et al. 2020). To show their real dedication and sustainability to their stakeholders, more and more organizations are implementing GSCM participants (Ilyas et al. 2020). Academic research also shown that GSCM is crucial for getting a competitive edge (Qader et al. 2022). Reverse logistics, for example, has been proven. GSCM was utilized to increase the competitiveness of businesses. In a highly competitive global corporate climate, discovered that GSCM enhances firms in balancing economic and environmental performance. Factor et al. (2021) recently

came to the conclusion that people, even managers, have the capacity to seek for and produce knowledge. This concept will improve GSCM and increase market competition.

Regarding the expansion of academic study on GSCM and its effects, the analysis of the literature demonstrates (such as waste reduction, environmental reliability, and pollution reduction) that the GSCM is still only conceptually understood; it is not yet obvious from the new role of COVID-19 whether practices can contribute to the dynamic growth of organizational competitiveness (Junaid et al. 2022; Ilyas et al. 2020). For example, the impact of GSCM on a company's competitive edge has been disregarded. The GSCM falls short in addressing stakeholders' environmental concerns. The study of the prior literature uncovered a number of research gaps and constraints (Tunio et al. 2021) (Herrmann et al. 2021). First, instead of concentrating on it, we contend that more investigation is required. Good knowledge of the GSCM and the market's underlying process dynamism, such as the environmental management system toward organizational sustainability in the market. An embedded system and database are known as an EMS because it enables an organization to exchange resources and information with internal and external stakeholders of the firm in order to comply with rules (Soomro et al. 2021) (Liu et al. 2021). For the call of research, we responded to a search request by recommending that organization look into how GSCM can benefit rival businesses (Nikolaou et al. 2018; Imran et al. 2020). The use of EMS, according to academics, has the ability to boost organizational benefits like investment growth, attracting top talent, improving quality control, enhancing working conditions, luring new clients and suppliers, and lowering pollution. We concur that EMS not only has positive effects on the economy but also makes businesses more competitive. Therefore, it is crucial to think about how GSCM might boost your company's competitiveness by applying EMS at work during tumultuous times like as COVID-19. The mechanisms underlying EMS between GSCM and market competitiveness in aggressive organizational strategies have, however, received little study. Recognizing the EMS mediation process, we learned more about how a company may manage its competitive advantage more effectively during challenging times like COVID-19 in China.

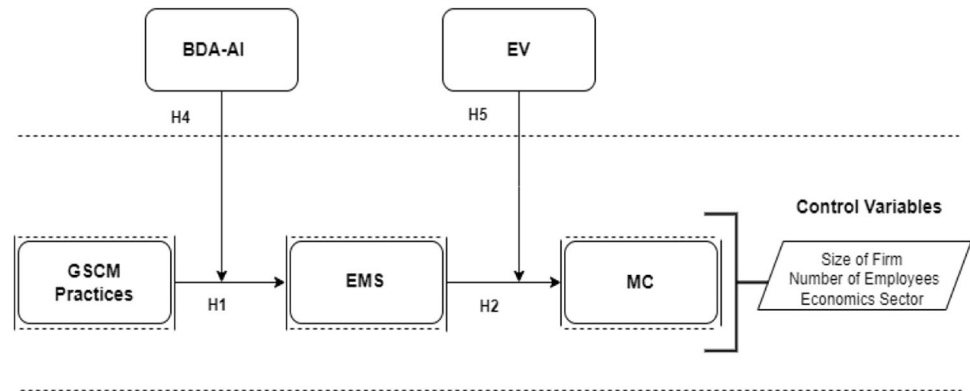
Second, the extensive use of ICT and big data is acknowledged in the prior literature as being strategically important to organizations for the creation of sustainable policies (Maheshwari et al. 2021). Big data is a term used to describe a range of information that is enormous and constantly expanding in terms of amount, diversity, speed, and correctness (Agbehadji et al. 2020). Data collection can be used by organizations to produce big data analytics and artificial intelligence (BDA-AI), which enables the transformation of data into informative information that

supports the supply chain and decision-making (Dubey et al. 2020a; Luqman et al. 2018). Therefore, future studies should expand on prior work on the connection between BDA-AI and the supply chain while also examining the role of BDA-AI. The BDA-AI GSCM looked at the BDA-regulatory AI's function between GSCM and EMS (Balan and Conlon 2018; Videgar et al. 2021). Studying the function of the BDA-AI is crucial since it is probable that the GSCM will be tightly controlled. BDA-AI is used by the company to support EMS inside the company. Furthermore, no studies examined how BDA-AI affected the regulation of the collaboration between GSCM and EMS. By studying how BDA-AI affects GSCM behaviors and related outcomes like market competitiveness, notably during COVID-19, the current study fills this information gap.

Third, the previous studies mostly focus on the coordinating role of intellectual capital (Khan et al. 2021), lockdown during COVID-19 (Benzidia et al. 2021), and green digital learning. As far as we are aware, no one has investigated how environmental visualization affects EMS and market competitiveness. The degree to which an organization informs stakeholders of its environmental policies and practices is known as environmental visibility (EV) utilization of the environment. The concept of environmental visibility is new in the literature on supply chains, although receiving a lot of academic attention in recent accounting studies. The rising demand for environmental data from stakeholders, in our opinion, will lead to a rise in the popularity of electric cars. As a result, in Phase 2 of the EV model function, we will look at EV regulation in addition to assessing the success of EV practices in the GSCM. In Fig. 1, Phase 2 of the VE is being taken into consideration by EMS since they feel that businesses are most successful when they share their real practices with stakeholders. This theory that EMSs are most effective when stakeholders trust organizational practices and they are compatible with what is actually disclosed is also supported by certain empirical data (Marshall et al. 2019). Therefore, the current study offers empirical proof of how the of EV well perform when GSCM and EMS are present in order to increase their marketability in COVID-19 in China.

Finally, we identified the following clear gaps in the GSCM literature: (a) the width and depth of study into the underlying mechanisms relating GSCM practices to market competitiveness are constrained; (b) it mostly focuses on visceral effects and does not offer comprehensive data, and these gaps represent unmet learning needs that have accumulated over time and hinder our current comprehension of how GSCM procedures affect businesses and their performance, particularly in pandemic scenarios like COVID-19. Beyond the overall effect of GSCM practices on organizational performance, take into account how GSCM practices link to EMS and how they relate to boundaries

Fig. 1 Hypothesized research model



given the unavoidable global spread of GSCM practices. The contemporary as it is market competition, EV circumstances, and BDA-AI must be controlled. We suggest two research problems in light of the gap and the requirement to close it: RQ1: How are market competition and the EMS affected by GSCM practices during COVID-19? RQ2: In the first and second stages of COVID-19's link between GSCM and market competitiveness, respectively, what roles do BDA-AI and EV play? This research made three novel contributions, which can be summed up as follows:

First, it broadens our comprehension of the constrained and sometimes skewed findings of earlier research on GSCM, which focused on outcomes like pollution reduction and supply efficiency. By particularly implementing GSCM procedures during COVID-19, an innovative EMS technique to boost market competition is provided by the current study. Second, make use of BDA-AI to enhance your EMS and learn more about how to obtain an advantage in the context of COVID-19. Here, recent research sheds light on how businesses can increase GSCM effectiveness by integrating BDA-AI. Last but not least, this study improves our comprehension of EV in relation to EMS and GSCM. GSCM and EMS that deal with environmental issues offer a combined database for the usage of information; thus, EV can assist in highlighting how important EMS is to maintaining market competitiveness in hazy circumstances like COVID-19 in China.

Literature review

Green supply chain management

The environment and the supply chain are emphasized in the existing literature while talking about GSCM (Cankurtaran and Beverland 2020; Canhoto and Wei 2021). A well-coordinated effort between company activities including purchasing, production, marketing, and transportation is necessary for GSCM (Herrmann

et al. 2021). To satisfy both end consumers and other stakeholders, GSCM needed a business plan that was in line with the supply chain process (Cankurtaran and Beverland 2020). Green business practices (GBP) are best adopted and used by companies to ensure that they are environmentally sustainable and to conserve resources in order to lower pollution, waste, and use of energy. The literature, for instance, identifies three key GSCM techniques: (1) reactive: “businesses devote less money to environmental management, mark reusable products, and to use ‘edge of pipe’ measures to decrease with their ecologic products; (2) proactive: “businesses begin to well before impending pollution regulations by recognizing a moderate resource management dedication to beginning product reuse and producing green products”; and at last, (3) integrative: businesses are involved in value seeking when they “integrate environmental operations like green procurement and ISO compliance as strategic efforts into their company strategy.” “At this time, businesses begin to anticipate prospective environmental regulations by making a small capital expenditure to start product recycling and developing green goods” (Baloch et al. 2022).

While the literature now in circulation covers the many effects of implementing GSMC practices, the debates around green supply chain management and market competitiveness are lacking and divisive. For example, the literature on GSCM shows varying outcomes about how effective GSCM is in organizations. Academics disagree on whether GSCM procedures increase profitability. Theoretical arguments for and against both of these ideas are presented by existing research. Academics, for example, contend that acquiring a competitive advantage may burden companies as GSCM methods raise the overall cost of business operations; serious worries have been raised concerning the loss of competitive advantage. According to a different study, GSCM procedures were more likely to increase operational effectiveness than financial success (Laari, Töyli et al. 2018), which was unable to demonstrate a link between GSCM and growing business profit.

However, because they have shown that GSCM can improve an organization's financial performance, researchers support its use. For example, stakeholder theory suggests that businesses create positive externalities when they internally impact both parties (stakeholders). Organizations are compelled by outside forces to implement green practices and decrease waste. According to the stakeholder theory, organizations' performance and financial health should increase when stakeholders are taken care of (Sarkis et al. 2011). Additionally, it has been demonstrated that a company can acquire a competitive edge by implementing environmentally friendly practices and obtaining the best stakeholder support (Nguyen and Adomako 2022). Few academics contest the idea that GSCM affects company profitability indirectly. For example, Feng et al. (2018) found that the organization's profitability is not directly impacted by GSCM. The primary objective of this essay is to raise awareness of administrative procedures and resource distribution, which might not instantly enhance financial performance.

Green supply chain management and COVID-19

The effectiveness of the GSCM is a topic of even greater discussion following the COVID-19 pandemic. Academic attention has been given to both the arguments in favor of and against the GSCM. Researchers contend, for example, that GSCM does not shield companies from the COVID-19 shock. They mature for a variety of causes. According to Laari et al. (2018), GSCM has no relationship with the organization's financial success and is not given preferential treatment by investors in uncertain economic times. Then, detractors seized the chance to assert that GSCM is ineffective during COVID-19. It was then brought up that the new difficulties brought on by COVID-19 might cause supply chain investments to drop. For example, a recent research by Baloch et al. (2021a) found that businesses commonly choose new environmental policies but abandon them when faced with unforeseen and new problems. Maintaining the impetus to engage in ecologically sustainable practices becomes a significant problem when financial resources are severely constrained (for example, as a result of COVID-19). Furthermore, the fact that COVID-19 is a social and health catastrophe rather than an environmental crisis underlies the GSCM's inefficiency. This might significantly shift a company's emphasis from social to environmental sustainability. For example, none of the participants from around the world listed the threat of infectious disease as one of the top five threats in the Global Risk Report 2020. Investors might think that since social and health issues will take precedence over environmental issues following COVID-19, this problem will be handled.

However, some contend that GSCM can shield businesses against disasters like COVID-19. According

to the stakeholder theory, companies who engage in CSR activities and establish environmental pledges with their stakeholders, particularly in times of crisis, may be treated more favorably. However, the findings imply that returns are modest but long-term (Talwar, Talwar et al. 2021). With the ability to quickly manage their supply chains and prevent sufferers due to raw material shortages, businesses investing in GSCM are encouraged to be more adaptable when coping with unexpected scenarios like COVID-19 (Canhoto and Wei 2021). According to recent research, the GSCM is a workable element of sustainable development that can support organizations and stakeholders over the long term while concentrating on abrupt environmental changes (Bhatia and Gangwani 2021), (Ikram et al. 2021). Additionally, according to Balan and Conlon (2018), GSCM enables businesses to enter new markets, enhance their reputations, and increase competition. Because of this, academics think that GSCM research should be thorough in both width and depth, especially in times of crises like COVID-19 and that it should be further investigated to boost market competitiveness and share. This study broadens our comprehension of the fundamental concepts underlying strategies like GSCM and EMS that increase market competitiveness (see Fig. 1).

Green supply chain management, environmental management system, and market competitiveness

The creators of ISO 14001 are aware of the critical role that an environmental management system (EMS) plays in any company's efforts to improve both its financial and environmental performance. EMS's mission is to highlight the value of an EMS to business and the fact that nobody can succeed in the market without a strong ISO 14001 EMS. For example, according to the literature now available, EMS (1) is an environmental policy committed to the prevention of pollution; (2) is a promotion of organizational planning, management, and monitoring policies to improve business; (3) increases management and employee adherence to environmental sustainability; (4) daily resources, such as training, for staff development; and (5) encourages the upkeep of EMS across the entire firm for the benefit of investors and other stakeholders. The European Commission (EC) created the well-known EMS instrument, which may help society transition to a more sustainable one (Fig. 2).

An environmental management system is a comprehensive system for gathering, analyzing, and communicating information about the professional environmental performance of businesses and stakeholders, which is made up of formal systems and databases (Johnstone and Hallberg 2020). The primary drivers behind the environmental management system's internal formal structure are resource conservation, waste management, training, and rapid reporting to senior management for the creation of the plan.

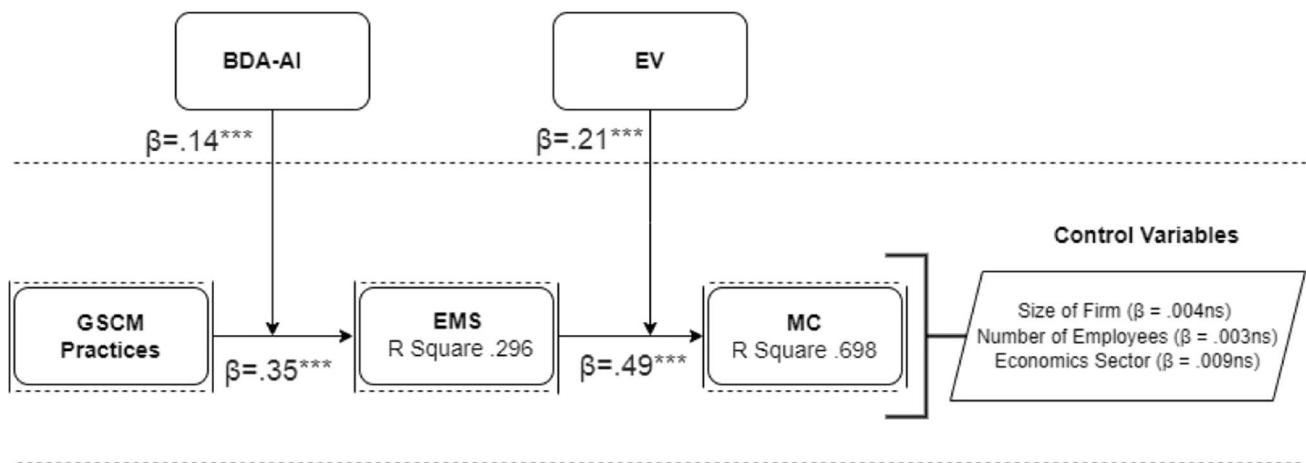


Fig. 2 Results of structural model

The company's annual report frequently includes this official information to profile it (Rodríguez-Anton et al. 2019). We concentrate on the connections between these systems, the environmental decisions made by businesses like GSCM, and their competitiveness in the market based on the descriptions given above. Competitiveness in the market is defined as "a market where numerous businesses compete against one another, but where one is outperforming the rest in terms of, for example, the sale of green products, growth in sales and market share, or exports. It is the element that enables us to develop high-quality products and services" (Cronjé and du Plessis 2020; Floričić 2020). As a result, the business makes more profit than its rivals (Benzidia et al. 2021). EMS is a method, not a particular performance measure of an environmental management system's main goal which is to make it easier for the organization to achieve its targeted environmental objectives. Both GSCM (described above) and EMS are applicable based on this methodology. An organization's capacity to oversee both GSCM and EMS has a beneficial effect on overall SCM and enhances performance (Kumar et al. 2020). A formal implementation of EMS, for example, has been shown to have synergistic impacts on organizational performance and is more likely to occur among GSCM adopters (Gualandris and Kalchschmidt 2016). They have not examined EMS's official affiliation with GSCM, though, and they have not discovered this association in emergency situations like COVID-19.

Recent research (Luqman et al. 2021) has concentrated on new organizational skills inside EMS integrated businesses, like employee representation and relationships with tools, knowledge exchange, and engagement, which can boost the market's competitiveness. Management may more effectively manage resources and operations with the help of GSCM and a strong EMS. Organizations can better apply their GSCM procedures with the use of an EMS.

For example, EMS makes it simpler for businesses to use GSCM techniques while managing organizational change and development (Ahmed et al. 2021), improving their financial stability (Canhoto and Wei 2021). Additionally, the usage of cross-functional resources and data inside an organization is made possible by EMS, which can aid in the management of environmental challenges (Johnstone and Hallberg 2020). The implementation of EMS, for example (Johnstone and Hallberg 2020), has been proposed to be able to save resources. It could result in small and medium-sized businesses (SMEs) having excellent financial stability (Gualandris and Kalchschmidt 2016). Make the case that the only strategy for enhancing organizational performance is a comprehensive EMS.

The literature that is now available discusses several market competitiveness factors. For example, strengthening an organization's market competitiveness can benefit from its reputation. A company's external reputation is enhanced through EMS certification, such as ISO 14001, eco-friendly business practices, and eco-innovation (such as new eco-compatible goods), which boosts its marketability and profits (Ikram et al. 2021; Gualandris and Kalchschmidt 2016). According to the aforementioned analysis and suggestions, integrating GSCM with a formal EMS can boost an organization's competitiveness in the market, notably during COVID-19, in addition to assisting it in managing its limited resources. We think it will contribute to us having more stable finances. Therefore, we speculate that:

Hypothesis 1. There is significant relationship between green supply chain management practices and environmental management system in organization.

Hypothesis 2. There is a significant relationship between EMS and market competitiveness in an organization.

Hypothesis 3. The relationship between green supply chain management practices and market competitiveness is mediated through EMS.

Big data analytics and artificial intelligence

The big data describes vast or intricate data sets that frequently exceed exabyte. This surpasses the limitations of conventional systems with regard to storage, processing, monitoring, decryption, and presentation (Gualandris and Kalchschmidt 2016). The amount of data is rapidly approaching zettabytes per year (Sun et al. 2020) and is expanding every day. Organizational change is being fueled by the IT revolution, consumer awareness, and globalization in today's highly competitive business. Rivalry inside firms has been replaced by competition between organizations and supply chain tactics. Supply chain experts manage enormous amounts of data in order to deliver integrated, effective, efficient, and agile supply chain systems (Benzidia et al. 2021). The expansion of data volumes and data kinds along the supply chain has sparked the need for systems that can quickly and intelligently evaluate large amounts of data. Implementing BDA-AI is the best practice for assisting organizations in obtaining valuable information from vast amounts of data and using that information to address supply chain issues (Maheshwari et al. 2021; Benzidia et al. 2021).

The function of BDA-AI in GSCM has recently come to the attention of researchers (Maheshwari et al. 2021; Toorajipour et al. 2021). The GSCM has emerged as a top concern for managers, legislators, and the general public as a result of growing environmental awareness of global warming, hazardous pollutants, and chemical spills (Sun et al. 2020). Two rising markets, India and China, are implementing digital technology like intelligent inspection equipment to monitor environmental hazards both within and across businesses in order to facilitate GSCM. Jiangsu Province in China, for example, employs smart devices to generate millions of distinct streams of unstructured data and collect environmental data in real time (Dubey et al. 2020b). In this situation, BDA-AI can assist in processing unstructured data to generate insightful conclusions. For instance, BDA-AI may assess dynamic data on energy use and carbon emissions in real time to support manufacturing process optimization with the aim of lowering emissions and conserving energy. Anecdotal information suggests that few companies use BDA-AI in their GSCM procedures. In its 2016 corporate social responsibility report, “Beijing Tiantan Biological Products Company,” for example, noted that “real-time monitoring bolsters the administration and oversight of energy use using technologies like big data.” BDA-AI enables businesses to produce insightful data to advance their environmental practices (Baloch et al. 2020; Agbehadji et al. 2020). Additionally, BDA-AI methods help firms take

advantage of green product breakthroughs while reducing resource waste and carbon emissions (Benzidia et al. 2021; Daddi et al. 2021). We think that using BDA-AI can help enterprises apply GSCM procedures more effectively and boost EMS performance. Consequently, we suggest the following presumptions:

Hypothesis 4. BDA-AI strengthens the relationship between GSCM and EMS practices so that it is stronger during COVID-19 when BDA-AI usage is high (relatively low).

Environmental visibility

It is recognized that environmental information is available to stakeholders and that this has an impact on business operations and stakeholder expectations. An important step in starting a conversation to take the requirements of stakeholders into consideration and make decisions accordingly can be the disclosure of environmental information to stakeholders. Furthermore, stakeholders must consider a variety of profiles when reacting to the many constituencies engaged in corporate performance (Gerged 2021). Businesses are raising the general public's knowledge of environmental issues as a result by implementing strategies like “CSR reporting,” “sustainability,” and “triple bottom line.” These reports make a case for visibility and transparency to a wide range of stakeholders, including shareholders, governments, NGOs, trade unions, clients, and other stakeholders (Baierle et al. 2022). More generally, the body of research implies that more stakeholder inclusion will help organizations using EV (Baloch et al. 2021b) (Qian and Chen 2021). Stakeholder input is included in environmental disclosure, which helps us improve our internal environmental policy (Qian and Chen 2021) and boosts market competitiveness (Floričić 2020). Extensive research has demonstrated that addressing certain stakeholder issues can have a broader impact and that environmental disclosures affect profitability by enhancing corporate reputation. The information that EV conveys should be consistent with that of the institutions established to solve sustainability challenges. If the reported policy and the actual policy diverge, trust in disclosure relationships may be jeopardized. For example, stakeholders can simply spot inconsistencies among actual and visible policies, such as ecological observes and environmental management systems, making it difficult for organizations to implement these visible policies within their organizations, according to researchers (Velte et al. 2020). Therefore, by formalizing EMS and organizational reputations, mistrust in this relationship can be eliminated. Following that, EVs are probably going to make the business more competitive in the market. Compared to other company operations like production and distribution, EVs have a more positive impact on

environmental sustainability within organizations. Companies should do better by sharing transparent data about their environmental performance across their supply chains using EV best practices. Transparency in disclosure, for example, entails informing our stakeholders, assessing and controlling our supply chain operations and activities, and putting our EMS practices into practice. As a result, it is consistent with Marshall et al.'s (2019) study using the transparency approach; it is suggested that EV can improve the connection between EMS and competitiveness of market.

Hypothesis 5. EV moderates the relationship between EMS and market competitiveness, and the relationship is stronger when EV is higher (vs. lower) in COVID-19.

Figure 1 shows the hypothesized research model.

Research methodology

The objective of this research is to assess the EMS mechanisms via which GSCM operations have an effect on company competitiveness. Additionally, we looked into the regulator functions that BDA-AI and EV each played in the first and second phases of the suggested model. We surveyed “Chinese people” to gather primary data for our evaluation of the proposed paradigm. Finding respondents from the food business that matched the inclusion requirements listed in the survey list required the use of Prolific, a web-based self-service online platform for data gathering. Additionally, WeChat was used in this procedure. The questionnaire was created based on current, verified evaluations from significant COVID-19 literature in the areas of logistics and GSCM. Previously launching research projects, operations and supply chain managers, academics, and scholars pre-tested our surveys a number of times. The survey questions were changed to better fit the topic of the study based on their input. To further explain the meaning of BDA-AI and EV, for example, we have contributed several examples and project glossaries to our research project. For supply and maintenance during COVID, we have designated the operations managers, supply chain managers, logistics managers, and intermediate managers. The current study focuses on organizational supply chain strategies and environmental sustainability. Respondents were given the assurance that their answers would be kept private and only used for research. Examples of target samples include lists of “Top Chinese Food Producers” and the “Chinese Food Producers Association.” Based primarily on the research background, we only hired individuals from the food industry in COVID-19. With a turnover of over, the food market will generate US\$1,259.00 billion in sales in 2022. The Chinese food market is anticipated to expand

by 9.49% yearly (CAGR 2022–2027). The Chinese food market appears to be in the lead (Statista, July 2022). For example, residents and other stakeholders are paying more attention to how food corporations are acting responsibly toward the GSCM and EMS. China has the fastest-growing foodservice market in Asia as well as the second-largest economy in world; the food sector is crucial in sustaining the needs of the entire nation during the COVID-19 lockdown (Coluccia et al. 2021; Coluccia et al. 2021). During the first wave of COVID-19, the cost of manufacturing and utilizing perishable goods increased, but flexibility in logistics and transportation ensured customers' access to stable prices. A total of 450 businesses were approached about taking part in the study, and 390 of them accepted, yielding an 86.6% response rate. However, only 360 replies were submitted, and 30 of those were disregarded for having insufficient data. The model is tested using the final, 330-person sample. Table 1 details the sample's demographics.

Measures

Green supply chain management

GSCM uses both internal and external GSCM practices as its two evaluation criteria. In line with the theoretical viewpoint, Longoni and Cagliano recorded internal and external GSCM responses using an 8-item scale, respectively. An example is “During COVID-19, to what extent has your company adopted the following practices [waste reduction practices] in its manufacturing processes over the past year?”

Table 1 Respondents' profile (%)

Gender	Male	60
	Female	40
Age	20–30 years	38
	31–40 years	37
	41–50 years	25
	51–60 years	20
Qualification	Graduate	54
	Postgraduate	40
	Doctorate	6
Experience	5 years	15
	6–10 years	35
	11–15 years	22
	16–25 years	18
	Greater than 25 years	10
No. of employees	Less than 100	70
	101–250	30
Business type	Public industry	15
	Private industry	75
	Global industry	10

An example would be “during COVID-19”. The extent to which your organization has used the following practices in its manufacturing processes over the past year [sustainability-based supplier selection (for instance, eco-friendly technology and initiatives). They were all scored using a Likert scale of 1 to 5 points: “(1) not at all, (2) somewhat, (3) quite, (4) a lot, (5) very much.”

Environmental management system

With the help of a 15-item reconciliation scale from a reference, organizational EMS is evaluated (Sroufe 2003). “How much is the corporation involved in the activities of the environmental management system with the following stakeholders? [Officially defined EMS procedures]?” is an example project. Five points were used to catch each object, similarity scale: (1) not at all, (2) a bit, (3) quite, (4) a lot, and (5) very much.

Environmental visibility

Based on respondents’ perceptions of how much information an organization shared about their environmental activities, EV was calculated. EV was recorded using a scale of 10 items that was modified. A typical question asked, “How accessible is environmental information to the following stakeholders [suppliers] during COVID-19.” Five points were used to catch each object, Likert scale: (1) not at all, (2) a bit, (3) quite, (4) a lot, and (5) very much.

BDA-AI

A 6-item scale adopted by Benzidia et al. (2021) is used to measure BDA-AI. “How much big data analytics: information on artificial intelligence can be found at adopted at the time of organizational integration process” is an example project. To make better decisions, think about applying sophisticated analytical techniques (such simulation, optimization, and regression). On a Likert scale of 1 to 5, with the following options, all responses were recorded: (1) strongly disagree, (2) disagree, (3) neutral, (4) agree, and (5) strongly agree.

Market competitiveness

Evaluation of subjective measurement modification for market competitiveness was highlighted in Daddi et al. (2021). The project’s five-item scale is designed to gauge how competitive the market is. The level of market rivalry your business encountered in COVID-19 is discussed in the following paragraph. As an illustration, the simpler access to finance markets is brought on by lesser environmental risk: (1) strongly disagree, (2) disagree, (3) neutral, (4) agree, and (5) strongly agree.

Common method bias (CMB)

The data collected is self-reported and is influenced by CMB (MacKenzie and Podsakoff 2012). Deal with prospective CMB risks. First, in line with earlier studies (Luqman et al. 2021), during the initial stages of data collection, anonymization, and item decoding, we employed the Harman single-factor test to rule out CMB. Only 28.47% of the overall variance, or significantly less than 50%, could be attributable to a single factor, according to the data (Table 2).

Third, determine CMB issues from the data using the remaining rules and the CFA marker variable technique. Confirmatory factor analysis (CFA) is a process that consists of five steps: baseline model, method-C, method-U, method-R, and CFA with marker. There are no CMB issues in the data, according to the comprehensive CFA marker results (see Table 3).

Results and findings

To validate the proposed model, we used a structural equation model (SEM) with AMOS (27 V). The covariance (CB-SEM) and variance-based SEM (VB-SEM) tests, both of which are supported by AMOS, were used by the researchers to effectively test their hypotheses (Luqman, Talwar et al. 2021). Because it is more forgiving in terms of sample size and is therefore best suited to verifying the theory, we utilized variance-based SEM, also known as (VB-SEM), to test our model. We put the structural model and model measurement to the test.

Measurement model

Using the methods and criteria offered, we assessed the measurement model’s accuracy and validity (Fornell and Larcker 1981). These experts advise using composite reliability (CR) and factor loading (confirmatory factor analysis) to verify validity, with outcome values exceeding the advised cut-off (0.7). Similar to this, AVEs should be utilized to determine the construct’s correlation and recommended cut-off value (0.5) for validity evaluation. Except for three items that we omitted from the analysis, all CFA values and CR in our situation are greater than 0.70, demonstrating that there is no convergent validity issue. Additionally, the validity tool developed by Fornell and Larcker (1981) was used to compare the correlations between latent components and AVE values in order to interpret the values. The discriminant validity indices for the component indicate quite outstanding validity, and the AVE of all factors is higher than the MSV and ASV. Table 4 below demonstrates the existence of discriminant validity according to the experts’ evaluation techniques (Fornell and Larcker 1981).

Table 2 Human's single factor

Total variance explained		Rotation sums of squared loadings			
Initial eigenvalues					
	Total	% of variance	Cumulative %	Total	Cumulative %
1	10.711	25.502	25.502	6.122	14.576
2	5.785	13.775	39.277	5.835	28.469
3	3.768	8.971	48.248	4.852	40.021
4	2.226	5.299	53.547	3.513	48.385
5	2.075	4.941	58.488	3.072	55.701
6	1.745	4.154	62.642	2.451	61.535

We also challenge the viability of the HTMT approaches put out by Altieri et al. (2015). Researchers today frequently use this method (Luqman et al. 2021). In order to obtain the appropriate level of validation, experts advised that the values should be lower than 0.85 at the time we were considering a unique tool (Altieri et al. 2015). The values from the HTMT findings are displayed in Table 5 as being at the recommended level (i.e., 0.85).

All factors have AVEs that are higher than their respective MSVs and ASVs, and the factor's discriminant validity indices indicate that its validity is generally quite good. Before applying path links between the latent variables of the inquiry, experts advise analyzing the model fit indices. The following model fit indices were created using the AMOS tool and evaluated using suggested cut-off settings (Fornell and Larcker 1981). The cut-off values and current model fitness values are listed in Table 6.

Structural model

Table 7 displays the outcomes of the structural stage. First, let us suppose that GSCM and EM in H1 have a positive association. Positive results and follow-up effects, such as GSCM-EMS compared to GSCM-EMS ($\beta = 0.34^{***}$; $SE = 0.06$; $p = 0.00$), support this idea. To bolster the results about the EMS megaproject design and market competitiveness, the following ideas are also put forth. We predict that SMEs during COVID-19 and their market competitiveness will have a favorable connection in the second half of the year. This hypothesis has been tested by promising findings, especially the economic viability of EMS ($\beta = 0.58^{***}$; $SE = 0.08$; $p = 0.00$). In order to achieve good relevance, H3 believes that the EMS will serve as a bridge between the GSCM and market competitiveness. Positive results, such as GSCM Indirect EMS market competition ($\beta = 0.18^{***}$; [bias-corrected confidence interval: drop of 0.20, rise of 0.27, $= 0.02$, $p = 0.00$]), show that the data supports this theory. In H4, we suggest that BDA-AI coordinate the interaction between GSCM and EMS for regulatory analysis objectives. This idea is supported by the fact that GSCM and EMS are positively regulated ($\beta = 0.14^*$; $SE = 0.03$; $p = 0.00$) (see Table 7 and Fig. 3). Together, the results of the moderation process imply that businesses with superior BDA-AI deployment options are more likely to have more EMS practices and higher BDA-AI ratings. The survey findings also revealed that respondents who utilized BDA-AI the most were more inclined to engage in SMU. The fifth and final hypothesis is that EV mediates the connection between market competitiveness and EMS. This idea is supported by a favorable adjustment between the competitiveness of SMEs and that of the market ($\beta = 0.47^{***}$; $SE = 0.02$; $p = 0.00$) (see Table 7 and Fig. 4).

Table 3 CFA with marker variable fit indices

Model	χ^2 (df)	CFI	RMSEA (90% CI)	LR of $\Delta \chi^2$	Comparison
CFA with marker	1929.8 (1040)	0.97	0.056 (0.061–0.75)		
Baseline	1875.4 (1040)	0.96	0.057 (0.079–0.087)		
Model-C	1859.407 (784)	0.93	0.058 (0.019–0.46)	27.195, df = 4, $p = 0.003$	Baseline
Model-U	1860.3 (742)	0.94	0.055 0.076–0.094	17.006, df = 42, $p = 0.064$	Method-C
Model-R	1866.1 (750)	0.93	0.054 (0.070–0.086)	16.7, df = 27, $p = 0.890$	Method-U

Discussion

This model creates suggested collaborations between GSCM, EM, EV, BDA-AI, and market rivalry by capturing an empirical snapshot of the Chinese food sector during COVID-19. This study supports the concept by showing how market competitiveness may be raised during times of crisis like COVID-19 by using strategies like GSCM, EM, EV, and BDA-AI. By concentrating on GSCM and BDA-AI deployment, businesses may enhance their environment and eventually increase their competitive edge. Additionally, as illustrated in Fig. 2, the study offers five hypotheses to investigate in multivariate analysis.

RQ1 during COVID-19 addressed how GSCM methods may boost EMS effectiveness. To try to find a solution to this question, we immediately looked into the interaction between GSCM and EMS (i.e., H1). The outcomes supported the worth of GSCM in terms of how it enhances EMS inside businesses (Flinchbaugh et al. 2016).

This result is in line with prior assertions that EMS will probably increase the efficacy of GSCM (Aslam et al. 2021). It was suggested in the second portion of the RQ1 that there is a direct relationship between EMS and market competitiveness (i.e., H2). The findings confirmed our expectations, demonstrating that introducing EMS in COVID-19 will increase the market's competitiveness. Many scholars have differing perspectives on the usefulness of formal EMS in the workplace, and these findings

have contributed to the ongoing debate (Gualandris and Kalchschmidt 2016).

This study demonstrates that the company's adoption of EMS will unquestionably increase its ability to compete in the market during crises like COVID-19. We also look at how SMEs (also known as H3) affect how GSCM and market rivalry interact. The idea that the implementation of GSCM inside an organization enhances EMS performance is supported by a mediation effect. It is therefore recognized that EMS has the capacity to utilize, operate, and deploy GSCM efficiently, boosting market competitiveness (Liu et al. 2021). The findings are backed not just by the EMS mediation strategy but also by stakeholder theory and other earlier studies (Nguyen and Adomako 2022). The findings also imply that by lowering stakeholder uncertainty during COVID-19, the adoption of GSCM and EMS can aid businesses in maintaining market competitiveness. Investors may decide to join the firm for its green and sustainable business style given the windfall that results from the company's erratic actions.

COVID-19 RQ2's Phases 1 and 2 are devoted to reducing the negative effects of BDA-AI and EV (i.e., H4 and H5) on the relationship between GSCM and market competitiveness. In order to respond to the first part of the question, we looked at how BDA-AI affected the interaction between GSCM and EMS. The reduced impact of BDA-AI on the link between GSCM and EMS was clearly confirmed by the experimental findings. When businesses employ BDA-AI

Table 4 Reliability and validity test results

		X	SD	CR	AVE	MSV	ASV	1	2	3	4	5	6
1	BDA-AI	4.34	2.22	0.86	0.61	0.18	0.08	0.78					
2	EMS	4.78	1.86	0.97	0.58	0.32	0.2	0.36	0.76				
3	EV	4.76	1.89	0.93	0.52	0.26	0.09	0.29	0.51	0.72			
4	GSCM-internal	4.55	1.93	0.96	0.78	0.32	0.14	0.42	0.55	0.23	0.89		
5	MKTC	5.16	2.22	0.96	0.83	0.25	0.12	0.28	0.48	0.15	0.48	0.91	
6	GSCM-external	4.12	2.11	0.87	0.63	0.000	0.000	−0.02	0.07	0.08	0.02	−0.03	0.78

Table 5 Heterotrait–monotrait (HTMT) test results

		1	2	3	4	5
1	BDA-AI					
2	EMS	0.63				
3	EV	0.568	0.296			
4	GSCM-internal	0.492	0.17	0.475		
5	MKTC	0.402	0.354	0.422	0.284	
6	GSCM-external	0.082	0.067	0.016	0.049	0.016

for decision-making, especially in times of crisis, the connection between GSCM and EMS grows depending on the good moderating effect BDA-AI has on this relationship (e.g., COVID-19). Businesses who employ BDA-AI are more likely to use the GSCM strategy and increase SME productivity. The findings regarding the use of BDA-AI as a mediator are based on a number of prior studies where experts used BDA-AI as a mediator to confirm its applicability to a variety of topics and dimensions outside of EMS and GSCM. The second section of QR2 shows that the relationship between EMS and market competitiveness is strengthened by electric automobiles. Companies can raise their level of market competition by disclosing taller EVs to interested parties. These findings are expected to result in an open and transparent exchange of environmental information with all relevant parties given the nature of VE. Gaining profitability quickly gives a business a competitive edge. Our results are in line with earlier studies that have shown that organizations that are transparent can gain the trust of their stakeholders and possibly enhance their financial success.

Implications, limitations, and future research

Theoretical implications

First, the current study contributes to our understanding of how EMS operates and how it relates to market competitiveness. We addressed stakeholder concerns regarding GSCM adoption and provided a more detailed explanation

of how GSCM adoption via EMS might improve market competitiveness. Regarding the success of GSCM methods in the company, previous research has produced conflicting findings, such as significant resource waste, rising business costs, and low profitability (Amankwah-Amoah 2020). The adoption of GSCM will, however, be more likely to be effective when deployed with EMS within the business, increasing the market competitiveness. We have supported this claim with empirical data.

Second, we observed that BDA-AI had a beneficial moderating impact on the GSCM-EMS connection. The statistical connections corroborated the claim made in a number of prior research that the use of BDA-AI improves stakeholder collaboration and organizational decision-making capacity (Maheshwari et al. 2021). The BDA-AI technique, for instance, has a favorable effect on supplier selection, according to recent research on engineering and technical management. During COVID-19, nobody looked at the function of BDA-AI in respect to GSCM and EMS, though. In our new contribution, we project that the COVID-19 problem will need the integration of GSCM and EMS into the internal workings of the food sector using BDA-AI technology. Furthermore, earlier studies support the fact that GSCM implementations are a real asset. For instance, with reference to COVID-19, we think that it is important to address and investigate the availability of big data in the food business as a resource that might help stakeholders feel less insecure.

Lastly, during COVID-19, we broaden the scope of the existing literature on EV to include stakeholders and the effects of EV on GSCM and EMS viewpoints. We

Table 6 SEM fit index results

Model test	CMIN/DF < 5	GFI > 0.9	AGFI > 0.9	NFI > 0.9	CFI > 0.9	RMSEA < 0.1
Measurement model	χ^2 (1929.8), df = 1040	0.95	0.96	0.93	0.97	0.059
	χ^2 /df = 1.85					
Hypothesis model	χ^2 (10.756), df = 08	0.97	0.96	0.93	0.91	0.27
	χ^2 /df = 1.344					

χ^2 chi-square, *DF* degree of freedom, *GFI* goodness-of-fit index, *AGFI* adjusted goodness-of-fit index, *NFI* normed-fit index, *CFI* comparative-fit index, *RMSEA* root mean square error of approximation

Table 7 Standard regression results

Variables	Environmental management system			Market competitiveness		
	<i>B</i> (se)	C.R	<i>p</i>	<i>B</i> (se)	C.R	<i>p</i>
Size of firm	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>	−0.004 (0.16)		0.66
Employee experience	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>	0.03 (0.09)		0.56
Economic sector	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>	0.009 (0.06)		0.17
Green supply chain management	0.34 (06)		0.00	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>
Environmental management system	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>	0.58 (0.08)		0.00
Analytics of big data and artificial intelligence	0.08 (0.07)		0.00	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>
Environmental visibility	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>	0.23 (0.08)		0.00
<i>R</i> ²	0.289			0.701		
Indirect effects				<i>B</i> (se)	UCLI-LCLI	
Indirect effect of GSCM on market competitiveness via EMS				0.18 (0.02)	0.20–0.27	
Moderated effects						
Green supply chain management * BDA-AI					0.14 (0.03)	0.11–0.18
Environmental management system * environmental visibility					0.47 (0.02)	0.33–0.72

experimentally examine the moderating role of EV to establish a link between EMS adoption and market competitiveness. According to our research, EMS alone will provide stakeholders the false impression about how to apply GSCM and EMS practices, which will have a negative impact on market competitiveness. Our results concurred with current research by Marshall et al. According to Marshall et al. (2016), GSCM and EMS implementation will not be successful unless stakeholders are not aware that they exist. We assessed the recommended study request during COVID-19 from the viewpoints of GSCM, EMS, and market competitiveness, and we added to the body of current literature addressing the continuing EV issue. As far as we know, no one has examined the EV links between EMS and market competitiveness.

Practical implications

First, our research shows that GSCM will aid the business in successfully implementing EMS and achieving market competitiveness. The manager needs to take into account EMS, which can be the ideal strategy in COVID-19 to remain competitive in the market and gain from GSCM. Second, the organization has to put in place a formal EMS because it has a direct impact on market competitiveness, which is connected to greater employment, the best conditions for workers, and market recognition for accomplishing goals. Our findings also showed that EMS-enabled firms profit more from the application of GSCM techniques. Policymakers should give certified EMS deployment great consideration, particularly in a time of uncertainty like COVID-19. It is a crucial step toward achieving market

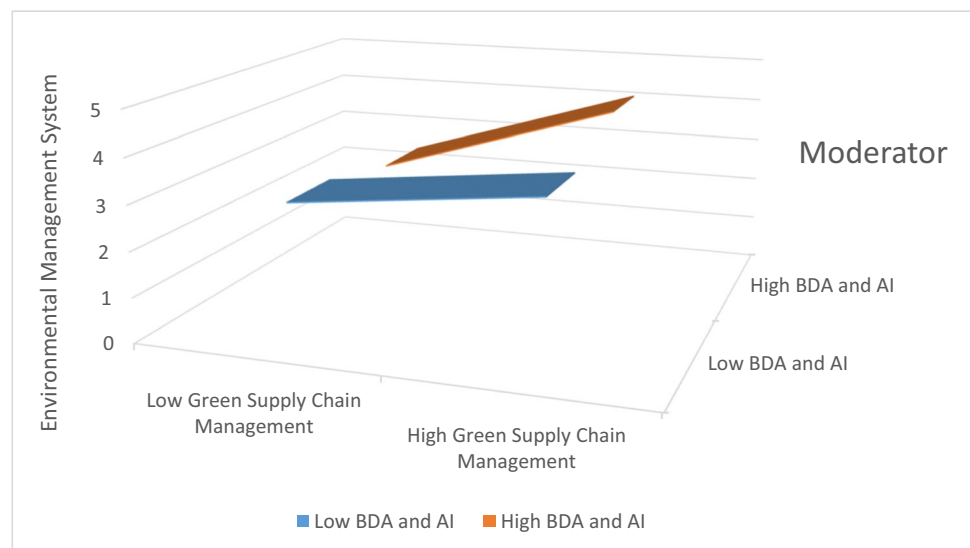
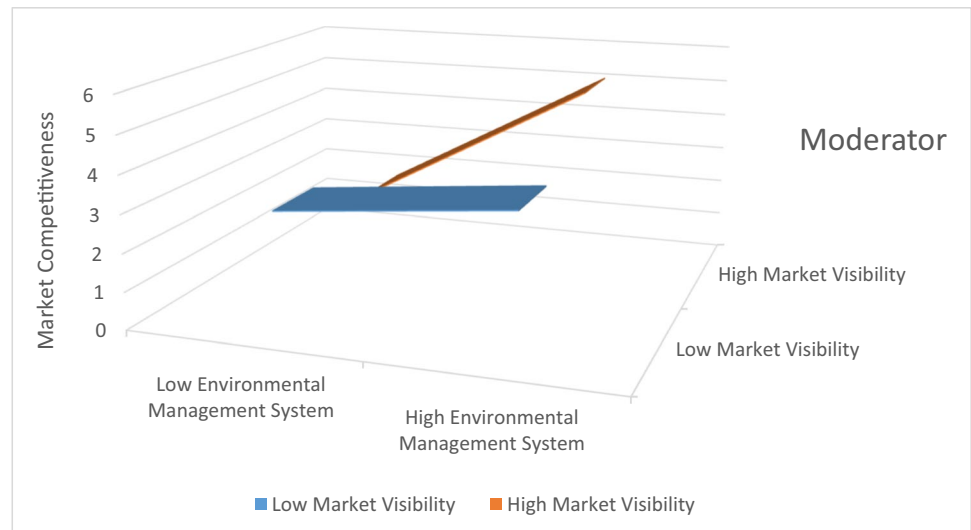
Fig. 3 Big data analytics and artificial intelligence moderating role

Fig. 4 Environmental visibility moderating role



competitiveness. Third, the study sheds light on how the practitioners actually use GSCM and BDA-AI, which will make an effect on EMS within the organization more effective. The practitioner ought to employ BDA-AI and extract pertinent information from the unprocessed data. It will further aid in enhancing sustainability and GSCM practices. Finally, we offer empirical support for our claim that communicating EV to stakeholders is more likely to improve EMS efficiency and, as a result, market competitiveness. Communication with stakeholders about the use of GSCM and EMS by firms will improve their reputation and trust in the market. The management should interact with the stakeholders and use their comments to enhance the EMS in order to prevent information from becoming unclear, which will ultimately benefit the companies.

Limitation and future research

Our study has merits, but there are limits on how far it can be used. For instance, the current study is based on data that was self-packaged and may have been affected by the CMB. We have ruled out all possibilities for the CMB data, however, to prevent compromising causality; we advise utilizing secondary data in addition to the original data. On the first and second levels, we then employ BDA-AI and EV as regulators. Future studies should concentrate on additional modifiers, such as stakeholders' transitive memory systems and managers' and employees' absorptive capacities. It opens up new avenues and presents academics and policymakers with fresh viewpoints on how to alter public perceptions of EMS. Third, the internal and exterior practices of the GSCM are accepted as its two-dimensional structure. To fully understand the influence on environmental management systems, more investigation is required to identify additional elements including green innovation, green goods, and green marketing. Additionally, we add the GSCM as a practice variable so that future research

may examine it in terms of experience and spot problems with GSCM implementation as a mismatch between real practices and demands. Fourth, we evaluate the effect of the GSCM on the EMS using the BDA-AI paper. Future studies would be more engaging if they concentrated on senior management's commitment to GSCM implementation as well as the leadership of logistics managers or SCs. These elements bring up fresh lines of inquiry, such as the impact of organizational traits on a firm's long-term strategy. Fifth, the fact that the data was gathered in China may restrict this study's capacity to be generalized. Finally, this study investigates how GSCM might enhance market competitiveness owing to SMEs in the context of BDA-AI and EV and discusses how future research could perform cross-country comparative studies to better understand GSCM practices during COVID-19. Given the breadth of our study, we did not concentrate on how consumers select environmentally friendly and sustainably produced goods, which can be more expensive than conventional goods. Determine which ones translate into market competitiveness if we are ready to pay more for sustainable items than cheaper alternatives (sustainable given likely cost).

Conclusion

The integration and coordination of SCMs that handle the demands of customers and suppliers have been adversely affected by the COVID-19 outbreak. The employment of GSCM and EMS, even during the pandemic, is stressed in recent study as a way to boost market competitiveness and offer long-term solutions. This study thoroughly examines whether EMS has an impact on the relevance of GSCM procedures and market competitiveness using SEM research and analytic approaches. This revelation has helped us

better understand how BDA-AI and EV are affecting how the GSCM process is managed. For managers, scholars, decision-makers, and professionals in the supply chain, current research offers crucial information. Even in times of crisis like COVID-19, businesses must implement formal EMS, BDA-AI, and EV systems to compete in the market.

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Author contribution Ghansham Das: data curation, methodology, and writing original draft. Shan Li: conceptualization, visualization, supervision, and editing. Raza Ali Tunio: data curation and visualization. Riaz Hussain Jamali: review and editing. Ihsan Ullah: writing, review, editing, and software. Kurukulasuriya Weerasinghe Tharindu Madushanka Fernando: methodology, data curation, and visualization.

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Data availability The data that support the findings of this study are openly available on request.

Declarations

Ethics approval and consent to participate We declare that we have no human participants, human data, or human tissues.

Consent for publication We do not have any individual person's data in any form.

Competing interests The authors declare no conflict of interest.

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